

EDUCATION

Sapienza University of Rome
Sep 2023 – Oct 2025

MS Economics and Finance – 110 cum laude

- Focus on quantitative methods and financial econometrics
- Thesis: Asset Pricing with Machine Learning
Supervisor: Prof. Giuseppe Ragusa

Sapienza University of Rome
Sep 2020 – Oct 2023

BS Economics

- Focus on Economic Theory
- Thesis: Dedollarization of the International Monetary System
Supervisor: Prof. Marco Di Pietro

ACADEMIC PROJECTS

Forecasting Macro Trends with Factor Models

Estimation of factors using PCA with missing data

- Applied to FRED-MD dataset (McCracken & Ng)
- Extracted 7 factors explaining 47.36% of variance up to March 2025
- Produced one-step-ahead forecasts for inflation and industrial production (RMSE/MAE evaluation)

Portfolio Construction

Sector-level portfolio optimization with Markowitz model

- Constructed sector portfolios and re-optimized them into a global efficient portfolio
- Documented dynamic portfolio weights (e.g. Health overweight in recent years)
- Evaluated stability through rolling-window estimation and out-of-sample checks

Volatility Forecasting

Comparison of econometric and ML models for Euro Stoxx 50 volatility

- Benchmarked GARCH as baseline model
- Multilayer Perceptron achieved superior out-of-sample forecasting performance
- Implemented hyperparameter tuning and validation splits for robustness

Identifying Macroeconomic Shocks

Structural identification of shocks via VAR and Local Projections

- Implemented Cholesky and sign-restriction approaches
- Produced impulse responses with confidence bands to assess macroeconomic dynamics

Statistical Models

Applied statistical learning to Human Development data

- Linear regression and Lasso with cross-validation for feature selection
- Classification via logistic regression, LDA, and K-nearest neighbors

Total Factor Productivity

Growth accounting using Solow residual and econometric analysis

- Implemented Solow model to estimate TFP across countries/sectors
- Data from Stehrer, R., A. Bykova, K. Jäger, O. Reiter and M. Schwarzhappel (2019)
- Computations and visualization performed in Excel

Wage Differences Analysis

Econometric study of wage determinants using synthetic dataset (Stata)

- Median wage comparisons by gender, education, and country
- Regression analyses with log(salary) as dependent variable
- Explanatory variables: education, gender, marital status, age, contract type, occupation, country
- Confirmed expected correlations (e.g. education, occupation) and observed wage gaps (e.g. gender, country)

Central Bank Policy: Federal Reserve Case Study

Comparative analysis of monetary policy tools and transmission mechanisms

- Reviewed Federal Reserve policy instruments: IORB, open market operations, repos/reverse repos, discount window, reserve requirements
- Analysed nontraditional tools used during crises (2008, 2020) and their impact on market functioning
- Studied interest rate corridor system (2021–2024) and compared to euro area operational framework

PERSONAL PROJECTS

Mean-Reverting Strategy in Euro FX Markets

Empirical analysis of euro exchange rates and short-term interest rate spreads

- Designed mean-reverting indicators using rolling Z-scores and volatility measures (ATR)
- Assessed performance of strategies on EUR/USD, EUR/GBP and EUR/JPY under different monetary policy regimes

Contrarian Strategy in Euro FX Markets

Statistical study of short-term reversal behaviour in euro currency pairs

- Detected contrarian signals during low-volatility phases in EUR/USD exchange rate
- Linked results to monetary policy cycles and volatility regimes in euro interest rate markets

SKILLS & INTERESTS

Technical Skills

- Python (NumPy, pandas, scikit-learn, statsmodels, matplotlib), R, Stata
- Econometric and statistical modelling: VAR, Local Projections, PCA/Factor Models, GARCH, ARIMA, Lasso & Regularization
- Data analysis and visualization; experience with large macro-financial datasets

Leadership & Teamwork

- Strong communicator and listener; able to present technical results clearly to diverse audiences
- Experienced in teamwork within multicultural and high-pressure environments
- Resilient, adaptable, and proactive in problem-solving

Languages

Italian (native), English (fluent), Spanish (intermediate), French (intermediate)

Interests

Free-diving, literature, soccer (both playing and watching), quantitative finance projects and strategy backtesting